

Big Mountain Resort Ticket Pricing Report

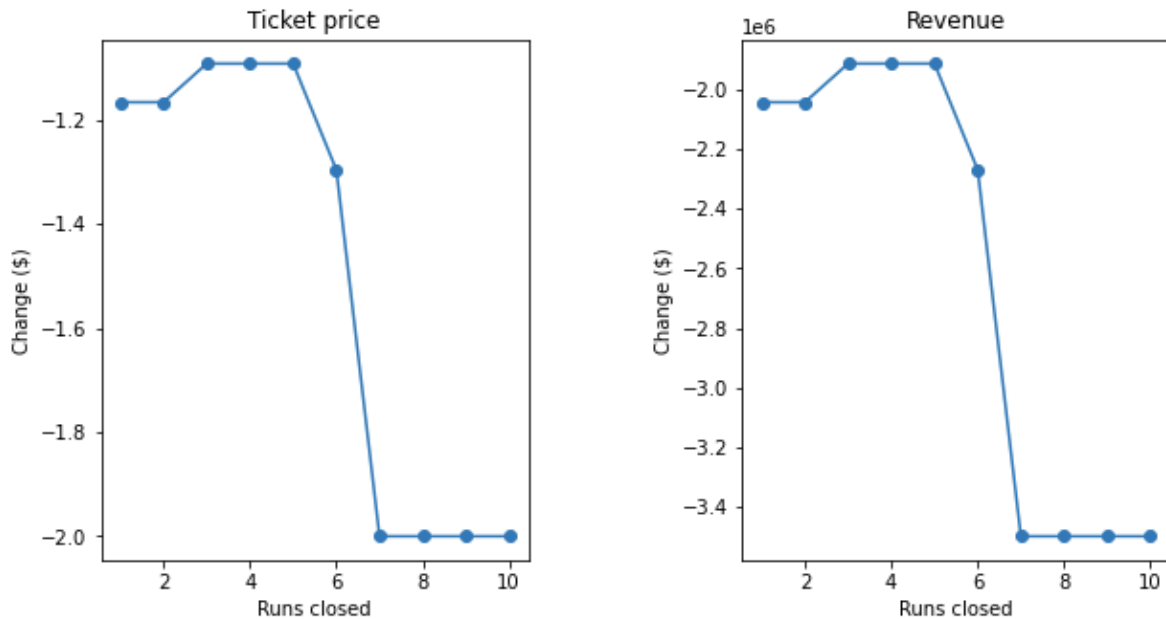
The executives of Big Mountain Resort are unsure how they should price their resort tickets in relation to other resorts and have asked us for strategy assistance. Due to the increased operation costs of installing a new lift (\$1,540,000), management wants to be able to offset these costs by determining if they can increase ticket prices as well as reducing costs in specific areas of the resort.

In order to figure out how Big Mountain Resort could offset the cost of installing a new lift, we utilized a data set that contained information about resorts across America – some of the data points include ticket prices for weekdays and weekend, how many chairs, and skiable terrain area. We have this data for about 330 resorts. But after diving deeper into the data set and cleaning it, 278 resorts remained due to missing data.

After cleaning the dataset, some exploratory data analysis was performed in order to determine what data points had strong relationships with ticket pricing. What we were able to uncover was that fastQuads, runs, total_chairs and snow making were some of the features that correlated the most with ticket pricing.

Once we were able to identify what features correlated the most with ticket price, we then created a model to predict what the ideal ticket price would be for Big Mountain Resort. Specifically, we built a linear regression model and a random forest model. The latter proved to be best as it showed that Big Mountain Resort could charge \$92.39 as opposed to the original price of \$81.

Additionally, once the model was deployed, there were four scenarios provided by the Big Mountain Resort management team to see what would be fruitful for the resort. The first scenario took a look at what potentially closing 10 runs would do to ticket price and revenue.



As you can see, closing just a single run would reduce the ticket price and reduce revenue for the resort. You can also see that if you close up to 5 runs the ticket price actually and revenue go slightly back up. If the resort had to close runs, they should close up to 5. Anything more would net a significant loss for the resort.

The next scenario was seeing what adding a run, increasing the vertical drop by 150 feet and installing a new chair lift would do to the ticket price. When we ran the model, the result below occurred.

This scenario increases support for ticket price by \$1.31
Over the season, this could be expected to amount to \$2300926

We can see that all these additions would be beneficial because we can raise the ticket price by \$1.31 and earn around 2.3 million dollars over the season. This would help offset the cost of installing a new chair to the resort.

The third and fourth scenario dealt with increasing the amount of snow making (in addition to scenario 2) and increasing the longest run/snow making, respectively. Scenario three did not increase the ticket price any more than scenario two and then scenario four was 0.

So in conclusion, Big Mountain Resort should only implement the second scenario. This is because it would increase the ticket prices by \$1.31 and generate potentially around 2.3 million dollars for the resort. This would be enough to offset the additional operating costs of \$1,540,00 from adding a new chair to the resort.